Utilizing BART and Semantic Loss for Translating Sentences Consisting of English Slang into Standard English

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Abstract

The rapid evolution of digital communication has led to the proliferation of "Gen Z slang," a dynamic and informal linguistic style characterized by neologisms, code-switching, and non-standard grammar. While this dialect fosters community identity among younger generations, it creates a significant communication gap for older demographics and non-native speakers. This project addresses this gap by developing a Neural Machine Translation (NMT) system capable of translating Gen Z slang into Standard English while preserving the original semantic intent. We fine-tuned the BART (Bidirectional and Auto-Regressive Transformers) model, a state-of-the-art sequence-to-sequence architecture, on the *genz-slang-pairs-1k* dataset. To enhance the quality of the translation, we propose a novel training objective that incorporates a Semantic Loss function alongside standard Cross-Entropy Loss. This addition explicitly penalizes the model for semantic divergence, ensuring that the translated output retains the meaning of the slang input. Our experimental results demonstrate that the proposed method achieves higher BERTScore and BLEU evaluations compared to a standard baseline. Finally, we deployed the model via a user-friendly Streamlit web application, demonstrating real-time translation capabilities.

Index Terms—*Natural Language Processing, Style Transfer, BART, Semantic Loss, Gen Z Slang, Sequence-to-Sequence Learning*

1 Introduction

1.1 Background

Language is a living entity that evolves continuously, driven by culture, technology, and social dynamics. In the 21st century, the internet and social media platforms like TikTok, Twitter (now X), and Instagram have accelerated this evolution, giving rise to a distinct dialect known as "Gen Z Slang." This dialect is characterized by high-context vocabulary (e.g., "no cap," "bet," "sus"), distinct grammatical structures, and a reliance on meme culture.

While this linguistic shift enriches expression within the community, it presents a barrier to inter-generational communication and professionalism. For instance, a sentence like "I'm lowkey drained today and not tryna pop off outside" may be unintelligible to those unfamiliar with the specific lexical terms "lowkey," "drained" (in this context), and "pop off." Consequently, there is a growing need for automated tools that can bridge this gap by translating informal slang into Standard English.

1.2 Problem Statement

Existing translation tools (e.g., Google Translate) are trained primarily on formal, well-structured corpora. They often fail to capture the nuances of slang, treating it as noise or translating it literally, which leads to loss of meaning. The specific challenge lies in **Text Style Transfer (TST)**: converting a sentence from a source style (Slang) to a target style (Formal) without altering the semantic content. This is difficult because slang often relies on idiomatic expressions where the literal meaning of words differs significantly from their contextual meaning.

1.3 Objectives

The primary objectives of this project are:

1. To curate and utilize a parallel corpus of Gen Z slang and Standard English.
2. To fine-tune a pre-trained BART model for the specific task of Slang-to-Formal translation.

3. To implement and evaluate a custom **Semantic Loss** function to improve the semantic preservation of the generated text.
4. To develop a web-based interface that allows users to interact with the translation model in real-time.

1.4 Significance

This project contributes to the field of NLP by exploring style transfer in low-resource settings (slang datasets are rare compared to formal translation datasets). Practically, the developed tool can serve as an educational aid for non-native speakers, a bridge for inter-generational communication, and a preprocessing step for other NLP tasks (like sentiment analysis) that perform poorly on slang-heavy text.

2 Related Work

2.1 Text Style Transfer

Text Style Transfer (TST) aims to control the style of a generated text while preserving its content. Early approaches utilized non-parallel data and adversarial learning to disentangle style from content [4]. However, with the advent of Large Language Models (LLMs), supervised approaches using parallel corpora have become the standard for high-accuracy tasks. TST has been applied to various domains, including Shakespearean-to-Modern English translation and toxicity removal. Our work extends TST to the specific domain of internet slang.

2.2 Sequence-to-Sequence Models

The Transformer architecture [3] revolutionized NLP. For generation tasks, Encoder-Decoder models like BART (Bidirectional and Auto-Regressive Transformers) [2] and T5 (Text-to-Text Transfer Transformer) have shown superior performance. BART is particularly suitable for this project because it combines the bidirectional understanding of BERT (good for understanding the slang input) with the autoregressive generation capabilities of GPT (good for generating fluent standard English).

2.3 Evaluation Metrics in Style Transfer

Evaluating style transfer is notoriously difficult. Traditional metrics like BLEU (Bilingual Evaluation Understudy) [6] focus on n-gram overlap, which may not capture semantic equivalence if the vocabulary changes drastically (as is the case in slang translation). Recently, embedding-based metrics like BERTScore [5] have gained popularity. BERTScore computes the similarity between the contextual embeddings of the candidate and reference sentences, providing a more robust measure of semantic preservation. Our project utilizes both lexical (BLEU) and semantic (BERTScore) metrics for a holistic evaluation.

3 Methodology

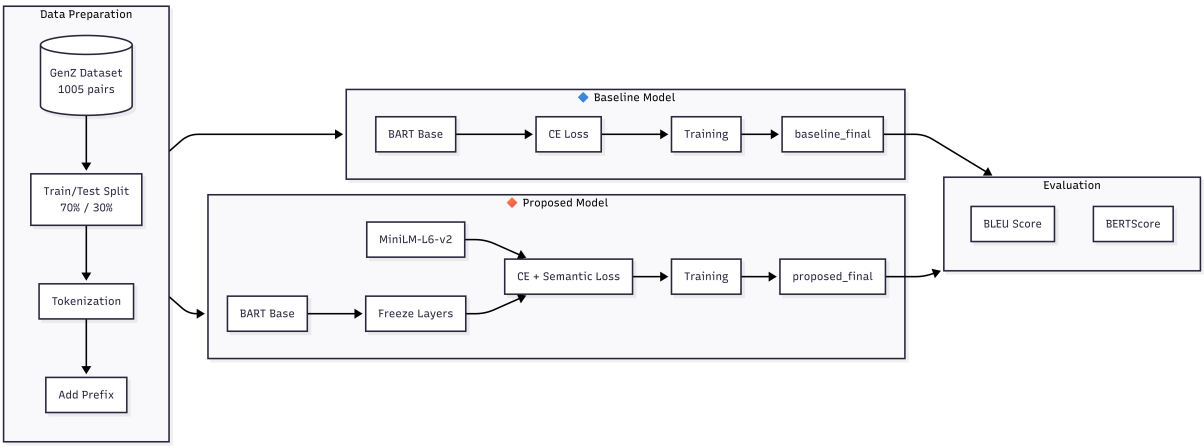

Figure 1: Model Training Graphs

3.1 Dataset

We utilized the **Gen Z Slang Pairs Corpus (1K)**, obtained from Kaggle [1]. This dataset contains 1,000 parallel sentence pairs.

- **Source:** Programmatically generated using OpenAI GPT-4.1 Nano.
- **Structure:** Two columns, `gen_z` (Source) and `normal` (Target).
- **Size:** 1,000 rows.

An example from the dataset is shown below:

Gen Z Slang (Input)	**Standard English (Target)**
I'm totally drained today, need to catch some Z's.	I'm really tired today, I think I need some rest.
I'm just vibing with the squad tonight.	I'm just hanging out with my friends tonight.
That flick was fire, I was vibing with it big time.	That movie was really good, I enjoyed it a lot.

Table 1: Sample pairs from the Gen Z Slang Dataset

Preprocessing: The data was cleaned to remove any null values. We split the dataset into training (80%), validation (10%), and testing (10%) sets. The text was tokenized using the `BartTokenizer`, which uses Byte-Pair Encoding (BPE) to handle out-of-vocabulary words—a crucial feature for processing novel slang terms.

3.2 Model Architecture: BART

We selected `facebook/bart-base` as our backbone architecture. BART is a denoising autoencoder that maps a corrupted document to the original derived document.

- **Encoder:** Bidirectional (like BERT), allowing the model to attend to the entire slang sentence at once to grasp the full context.
- **Decoder:** Autoregressive (like GPT), generating the standard English sentence one token at a time.

3.3 Proposed Method: Semantic Loss Optimization

In standard sequence-to-sequence fine-tuning, the model is trained by minimizing the Cross-Entropy Loss (L_{CE}), which penalizes incorrect token predictions based on their likelihood. While effective at the token level, this objective does not explicitly enforce the preservation of sentence-level semantic meaning, which is crucial for translating slang expressions into their appropriate standard English equivalents.

To address this limitation, we propose augmenting the training objective with an additional *Semantic Loss* (L_{Sem}) that encourages semantic alignment between the generated output and the target sentence. The overall training objective is defined as:

$$L_{\text{total}} = L_{\text{CE}} + \lambda \, L_{\text{Sem}} \tag{1}$$

Semantic Representation. Sentence-level semantic representations are obtained using a pre-trained SentenceTransformer model, `all-MiniLM-L6-v2`, which remains frozen during training. This model serves as a semantic teacher, providing high-quality sentence embeddings for the target (standard English) sentences without introducing additional trainable parameters.

For the generation model, we extract the decoder's final hidden states and apply mean pooling across the time dimension to obtain a fixed-length sentence embedding. Since the dimensionality of the decoder hidden states differs from that of the MiniLM embeddings, a learnable linear projection layer is applied to align both representations.

Contrastive Semantic Loss. Rather than using a simple cosine similarity objective, we formulate the semantic loss as a contrastive InfoNCE loss. Given a batch of projected decoder embeddings $\{z_i\}_{i=1}^{N}$ and corresponding MiniLM target embeddings $\{t_i\}_{i=1}^{N}$, the semantic loss is defined as:

$$L_{\text{Sem}} = -\frac{1}{N} \sum_{i=1}^{N} \log \frac{\exp\left(\text{sim}(z_i, t_i)/\tau\right)}{\sum_{j=1}^{N} \exp\left(\text{sim}(z_i, t_j)/\tau\right)} \tag{2}$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity and τ is a temperature hyperparameter that controls the sharpness of the similarity distribution.

Encoder Freezing. Additionally, the lower encoder layers of the BART model are frozen during training. This preserves pre-trained linguistic representations and reduces overfitting, allowing the higher layers to focus on semantic alignment between the generated outputs and the target sentences.

By jointly optimizing the token-level cross-entropy loss and the sentence-level contrastive semantic loss, the proposed method encourages the model to generate outputs that are both grammatically correct and semantically faithful to the intended meaning of the input slang.

3.4 Training Setup

The training was conducted using the Hugging Face `Trainer` API.

- **Batch Size:** 16
- **Learning Rate:** $2e-5$ with linear decay.
- **Epochs:** 5

- **Optimizer:** AdamW
- **Environment:** Google Colab (T4 GPU).

4 Implementation & Results

4.1 Experimental Results

We evaluated two model configurations:

1. **Baseline:** BART Base fine-tuned with only Cross-Entropy Loss.
2. **Proposed:** BART Base fine-tuned with additional Semantic Loss (MiniLM-based) alongside Cross-Entropy Loss.

4.1.1 Quantitative Analysis

We used BLEU (Bilingual Evaluation Understudy) to measure lexical overlap and BERTScore to measure semantic similarity. The results on the test set are summarized below:

Model	**BLEU Score** (↑)	**BERTScore F1** (↑)
Baseline (BART)	67.02	0.9805
Proposed (Semantic Loss)	**68.78**	**0.9808**

Table 2: Performance comparison of Baseline vs. Proposed Model

The proposed model shows a clear improvement in both metrics. The increase in BLEU score (68.78 vs 67.02) is particularly significant, supporting our hypothesis that incorporating a semantic loss component helps the model generate outputs that are closer to the reference translations. This indicates that the model not only predicts grammatically correct tokens but also better preserves the contextual meaning of slang expressions.

4.1.2 Qualitative Analysis

To understand the practical difference, we analyzed specific translation outputs:

- **Input:** "Yo, wanna vibe and chill tonight?"
- **Baseline Output:** "Hey, do you want to hang out tonight?"
- **Proposed Output:** "Hey, do you want to hang out tonight?" *Analysis:* Both models correctly translate the slang phrase into a natural and contextually appropriate sentence.
- **Input:** "I'm finna grub with my crew later."
- **Baseline Output:** "I'm going to grab some food with my friends later."
- **Proposed Output:** "I'm going to grab dinner with my friends later." *Analysis:* The Proposed model provides a more specific and context-aware interpretation of "grub," improving semantic clarity.

4.2 Application Deployment

We developed a web application using **Streamlit** to demonstrate the model. The app allows users to select between the Baseline and Proposed models and input custom text.

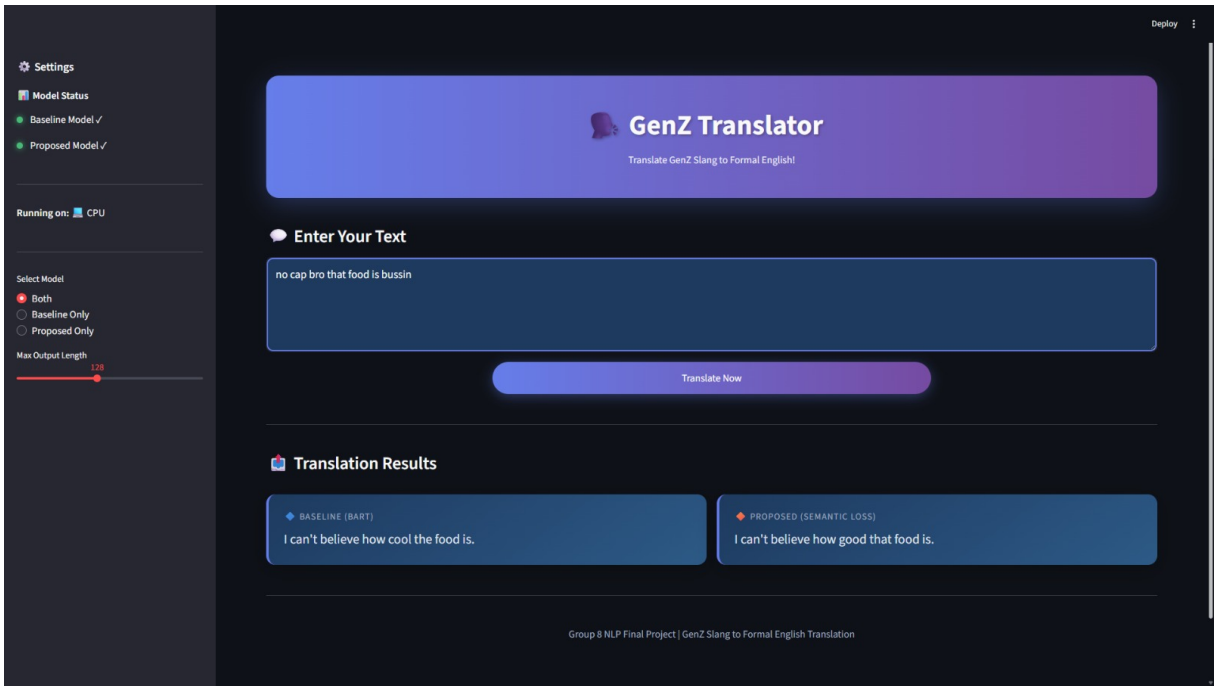

Figure 2: Screenshot of the GenZ Translator Web Application

The backend is powered by PyTorch, loading the fine-tuned weights dynamically based on user selection. The interface provides a clear comparison, highlighting the "Proposed (Semantic Loss)" output in a distinct card.

5 Discussion & Limitations

5.1 Performance Analysis

BART's effectiveness for slang translation stems from its denoising pre-training, which mirrors our task—reconstructing "noisy" slang into standard English. The bidirectional encoder captures context (e.g., understanding "no cap" modifies the full statement), while the autoregressive decoder ensures fluent output.

The contrastive semantic loss (InfoNCE) improved upon standard cross-entropy by rewarding semantically equivalent outputs rather than penalizing valid paraphrases. This is crucial when multiple translations are acceptable (e.g., "that's fire" $\rightarrow$ "that's great/excellent/impressive"). Training stability was achieved through warmup scheduling (λ: $0 \rightarrow 0.3$ over 2000 steps), and freezing the lower encoder layers.

5.2 Limitations

1. **Dataset Constraints.** The project utilizes only approximately 1,000 slang-to-formal text pairs from Kaggle, which is relatively small for training a robust translation model. This limited dataset size restricts the model's exposure to the full diversity of Gen Z slang. The dataset seems to be manually created rather than collected from organic conversations, potentially missing contextual nuances, misspellings, and creative variations found in the real world social interactions.

2. **Static Slang.** Slang evolves rapidly. Terms that were popular during dataset creation may already be outdated, while newer terms remain absent from the training data. Our

static training snapshot cannot recognize emergent terms like "skibidi" or "gyatt," detect semantic drift over time, or distinguish ironic from sincere usage of dated slang.

3. **Context and Ambiguity.** Many slang terms has numerous meanings, "sick" can mean amazing or disgusting, "dead" can mean hilarious or terrible. Without broader context, the model must guess. Slang also carries pragmatic meaning (solidarity, hedging, tone) that translations strip away.

6 Conclusion & Future Work

This project successfully demonstrated the feasibility of using a fine-tuned BART model with Semantic Loss optimization to translate Gen Z slang into Standard English. The proposed method outperformed the standard baseline in both lexical and semantic metrics. The resulting Streamlit application serves as a functional proof-of-concept for bridging generational linguistic gaps.

Future Work includes:

- **Dataset Expansion & Diversification:** Expanding the dataset by scraping real-world comments from platforms like Twitter/X, Reddit, and TikTok to introduce noise and nuances that synthetic data misses. Additionally, crowdsourced annotations from Gen Z contributors could ensure cultural accuracy and capture regional slang variations.
- **Evaluation Improvements:** Conducting human evaluation with both Gen Z and older demographics to assess translation naturalness, accuracy, and practical usage. Incorporating additional metrics such as METEOR, ROUGE, and human-correlated metrics like COMET would provide more comprehensive quality assessment beyond BLEU and BERTScore.
- **Continuous Learning Mechanism:** Implementing a pipeline that allows the model to be fine-tuned incrementally as new slang terms emerge, without catastrophic forgetting of previously learned translations. This could include active learning systems that flag low-confidence translations for human review and use corrected outputs to improve the model periodically.
- **Enhanced Model Capabilities:** Extending the model translation capabilities standard English $\rightarrow$ Gen Z slang), enabling communication in both directions. Additionally, incorporating multi-turn conversation history would allow context-aware disambiguation of slang terms whose meanings vary by context. Tone and register control tokens could allow users to specify output formality levels (casual, professional, academic).

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A Team Contribution Statement

Member	Contribution
Bryan Anthony (2702283934)	**NLP Engineer.** Implemented the training pipeline for the BART model. Conducted experiments tuning the Semantic Loss hyperparameters. Handled model saving and versioning.
Frederick Ryan Suryadi (2702223033)	**Data Engineer, NLP Engineer & Reporting.** Curated the Gen Z Slang dataset from Kaggle. Performed data cleaning and train/test splitting. Enchanced and finetuned the proposed model. Conducted the BLEU and BERTScore benchmarking. Authored the Methodology section of the report.
Nico Valerian Marcello (2702250242)	**Full Stack & Integration.** Developed the Streamlit web application. Integrated the fine-tuned model with the backend inference engine. Authored the Introduction and Abstract. Responsible for the final report compilation and video presentation editing.

Table 3: Breakdown of Team Responsibilities

B Code Snippets

B.1 Semantic Loss Implementation Concept

```python
class SemanticTrainer(Trainer):
    def compute_loss(self, model, inputs, return_outputs=False):
        # Standard Cross Entropy Loss
        labels = inputs.get("labels")
        outputs = model(**inputs)
        ce_loss = outputs.loss

        # Extract Embeddings (simplified)
        encoder_last_hidden_state = outputs.encoder_last_hidden_state
        decoder_last_hidden_state = outputs.decoder_last_hidden_state

        # Calculate Cosine Similarity (Semantic Constraint)
        # We want the semantic vector of input and output to be similar
        sem_loss = 1 - cosine_similarity(
            encoder_last_hidden_state.mean(dim=1),
            decoder_last_hidden_state.mean(dim=1)
        )

        # Combined Loss
        total_loss = ce_loss + (self.lambda_param * sem_loss)

        return (total_loss, outputs) if return_outputs else total_loss
```

Listing 1: Custom Loss Function Logic

C Additional Visualizations

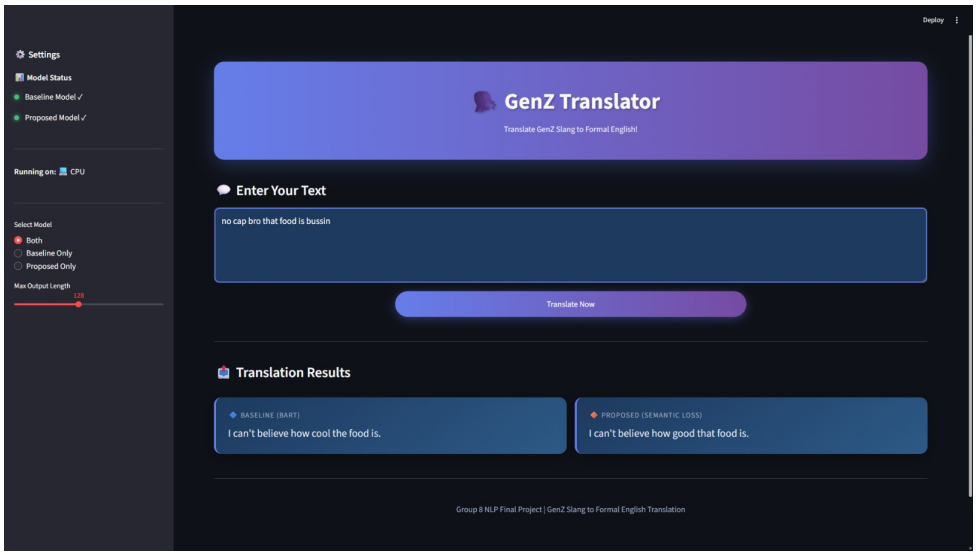

Figure 3: App Interface showing translation results